MATH2241 – Introduction to mathematical analysis (1st semester 2526)

Tutorial 7

---

1. (a) (Cauchy Condensation Test)Let $a_n \geq 0$ be a decreasing sequence. Prove that

$$\sum_{n=1}^{\infty} a_n \text{ converges } \iff \sum_{n=0}^{\infty} 2^n a_{2^n} \text{ converges.}$$

   (b) Let $p > 0$. Use (a) to prove that the p-series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges iff $p > 1$.

2. Find if $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^{1+\frac{1}{n}}}$ converges. (Hint: $2^n > n$).

3. Let $f : \mathbb{R} \to \mathbb{R}$ be a function and $c \in \mathbb{R}$. Suppose that $\lim\limits_{x\to c}(f(x))^2 = L$.

   (a) Use the $\varepsilon - \delta$ definition of limit to prove that $L = 0 \implies \lim\limits_{x\to c} f(x) = 0$.

   (b) Show by an example that $\lim\limits_{x\to c} f(x)$ need not exist when $L \neq 0$.

4. Suppose that $\lim\limits_{x\to c} g(x) = 0$ and $f(x)$ is bounded. Prove that $\lim\limits_{x\to c} fg = 0$.

5. Let $f(x) = \sqrt{x}$. Use the $\varepsilon - \delta$ definition to prove that f is continuous on $[0, \infty)$.

6. Let $f : \mathbb{R} \to \mathbb{R}$ be continuous at c. Suppose that $f(x) < 1$ for all $x \in \mathbb{R}\backslash\{c\}$, prove that $f(c) \leq 1$.

7. (Thomae)For a rational number $x = \frac{p}{q}$, where p is an integer, q is a natural number, and p, q are coprime, define $R(x) = \frac{1}{q}$. For an irrational number x, define $R(x) = 0$. Finally define $R(0) = 1$. Find all continuous point of $R(x) : [0, 1] \to \mathbb{R}$.

— End —